ATOM-SG Baseline Test

Version 6.0 - Parametric Approach

Generated: 2026-04-19

Instructions:

1. This test contains 40 questions total:

- 20 Word Problems

- 12 Geometry Questions

- 8 Data Interpretation Questions

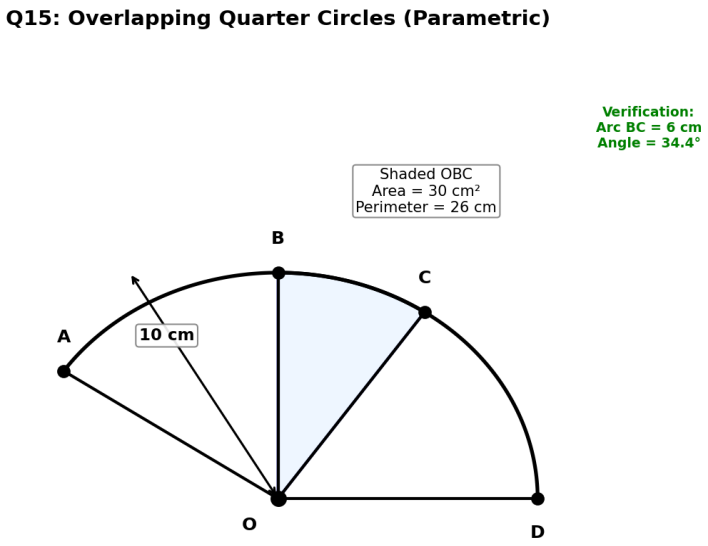
2. All diagrams use parametric rendering with variable-driven elements.

3. Show all working clearly.

Section 1: Geometry

Question 15 (5 marks)

The figure shows two overlapping quarter circles with centre O. Each quarter circle has a radius of 10 cm. The shaded part OBC has an area of 30 cm² and a perimeter of 26 cm. Find the area and perimeter of figure OABCD. (Take $\pi = 3.14$)

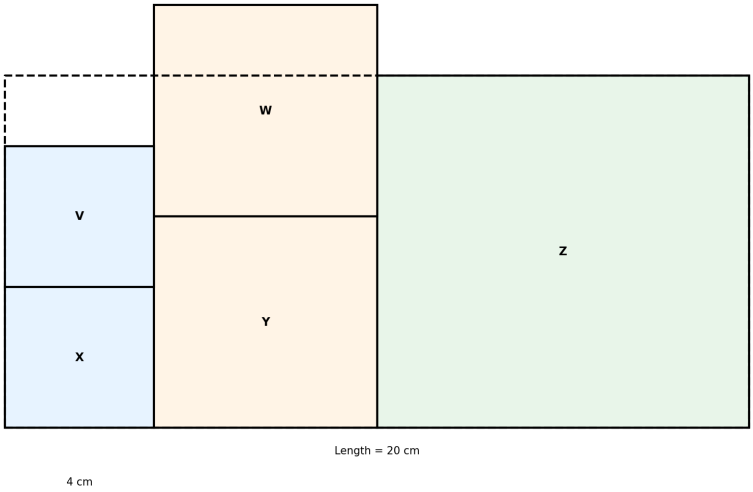


Answer: Area = _____ cm², Perimeter = _____ cm

Question 12 (4 marks)

The figure shows a rectangle made up of 5 squares. Square X has a side of 4 cm. Find (a) the side of square Y, (b) the length of the rectangle, (c) the number of 2 cm cubes that can be cut from the rectangle.

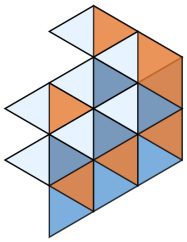
Q12: Five Squares in Rectangle



Question 19 (3 marks)

The solid below is made up of 7 identical cubes. The top view and side view are shown. Draw the front view of the solid on the grid below.

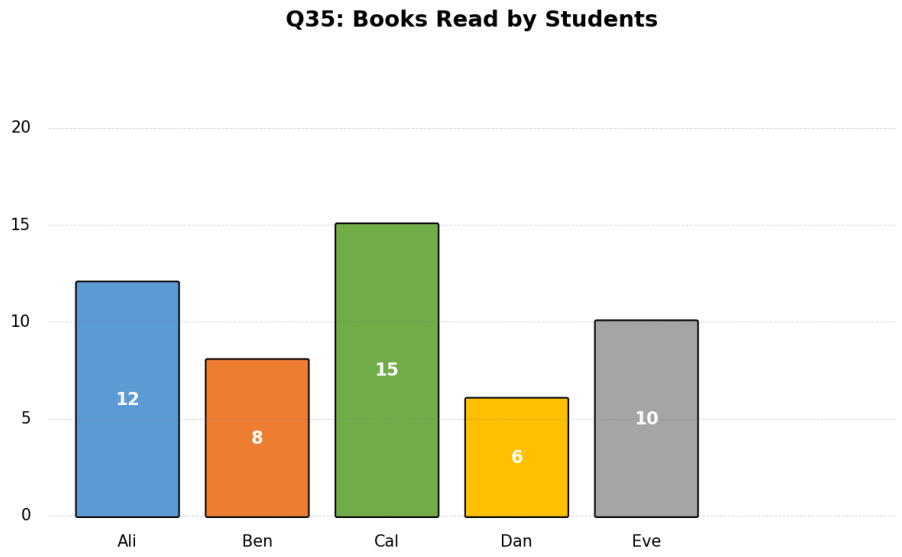
Q19: 3D Solid (7 cubes) - Isometric View



Section 2: Data Interpretation

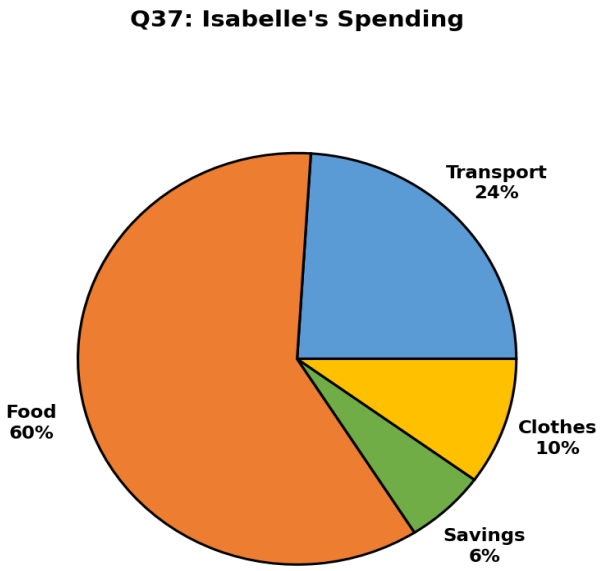
Question 35 (3 marks)

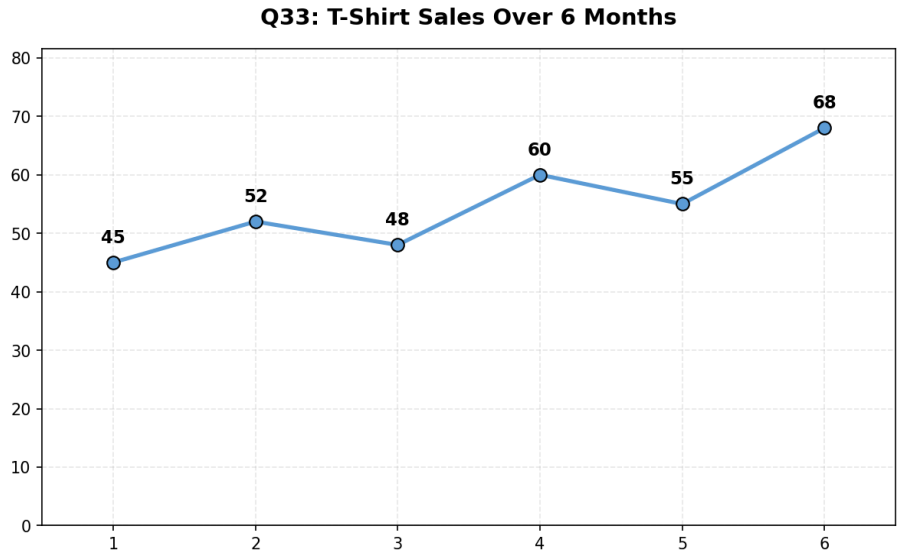
The bar graph shows the number of books read by 5 students. (a) Who read the most books? (b) How many books did they read in total? (c) What is the average number of books read?



Question 37 (3 marks)

The pie chart shows how Isabelle spent her $100 pocket money. (a) How much did she spend on food? (b) What fraction was spent on transport? (c) How much did she save?



Question 33 (4 marks)

The line graph shows the sales of T-shirts over 6 months. (a) What was the increase in sales from Month 1 to Month 4? (b) Calculate the average monthly sales. (c) Express the sales in Month 6 as a percentage of Month 1.